CIS 5450 Final Project Proposal

Station-Level Bike-Share Demand Forecasting
and Rider-Type Analysis in New York City

Qing Deng
qingdeng@engineering.upenn.edu

Zeyin Feng
zeyin@engineering.upenn.edu

Zhihao Huang
huan3134@engineering.upenn.edu

Jacob Zhu
jyz9029@engineering.upenn.edu

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1 Group Members and Responsibilities

- **Qing Deng:** Download and consolidate trip records, integrate weather and holiday data, and perform data cleaning.
- **Zhihao Huang:** Conduct exploratory data analysis, create visualizations, and interpret temporal and spatial patterns.
- **Jacob Zhu:** Develop and evaluate station-level departure and arrival demand forecasting models.
- **Zeyin Feng:** Develop the rider-type classification model and conduct hypothesis testing.

All members will contribute to interpreting results, integrating the final notebook, and preparing the presentation. Responsibilities may be adjusted as the project develops.

2 Data Source

Our primary dataset will be the publicly available New York City bike-share trip histories for **April–September 2025**.

- **Official source:** `https://citibikenyc.com/system-data`
- **Monthly downloads:** `https://s3.amazonaws.com/tripdata/index.html`
- **Data-sharing policy:** `https://citibikenyc.com/data-sharing-policy`

The current schema contains 13 fields, including ride ID, bicycle type, start and end timestamps, station identifiers and names, station coordinates, and membership category. These are real operational records rather than synthetic data or competition data. The data is publicly downloadable and licensed for analysis for lawful purposes, subject to the published restrictions.

We expect the selected period to provide substantially more than 50,000 trips after cleaning. We will verify this count and ensure that the station-hour analysis table also contains at least 50,000 valid observations.

We will join the trip data with NOAA weather observations, including temperature and precipitation, and a calendar of U.S. federal holidays. We will select an appropriate New York City weather station and document its coverage, missingness, and temporal resolution before modeling.

3 Objective and Value Proposition

The project will investigate how temporal, spatial, and weather-related factors relate to bike-share demand and rider behavior.

Our **primary objective** is to forecast hourly departures and arrivals at individual stations. Predicted net outflow, defined as predicted departures minus predicted arrivals, will help identify station-hours with potential inventory pressure and inform bicycle rebalancing priorities.

Our **secondary objective** is to characterize differences between member and casual rides. We will identify locations and times where casual rides with member-like characteristics are concentrated, providing exploratory evidence for selecting membership promotion locations.

Trip histories alone do not reveal historical bicycle inventory or dock availability. Therefore, we will forecast observed trip flows rather than directly predict empty or full stations. Without persistent rider identifiers or conversion outcomes, we will not identify individual customers or predict individual membership purchases.

4 Modeling Plan

4.1 Primary Task: Station-Hour Demand Regression

We will aggregate trips into a station-hour table and predict two outcomes for each station:

- Number of departures in the next hour.
- Number of arrivals in the next hour.

Candidate features include station identity or location, hour of day, day of week, holidays, lagged trip counts, historical demand summaries, and weather variables.

We will compare a historical station-and-hour average baseline with models such as Poisson regression and random forest regression. Performance will be evaluated using mean absolute error (MAE) and root mean squared error (RMSE), including performance during high-demand periods.

We propose the following chronological split:

- **Training:** April–July 2025.
- **Validation:** August 2025.
- **Testing:** September 2025.

Lagged features will use only information available before the forecast time. For prospective forecasting, weather inputs must also be available at that time. Any evaluation using realized future weather will be explicitly described as conditional forecasting.

This split evaluates prediction for subsequent months within the selected period, rather than establishing year-round generalization.

4.2 Secondary Task: Rider-Type Classification

The classification target will be whether a recorded ride belongs to a member or a casual user. Candidate features include trip timing, bicycle type, station locations, trip duration, and straight-line distance between endpoints.

We will compare logistic regression with a tree-based classifier. Evaluation metrics will include precision, recall, F1-score, and ROC-AUC.

Because duration and endpoint information are available only after a ride ends, this task will be framed as retrospective ride-pattern analysis. We will examine where and when casual rides assigned a high member probability are concentrated. These scores indicate similarity to observed member rides, not the probability of purchasing a membership.

5 Hypothesis Testing

Our primary hypothesis test will examine whether electric-bike usage differs between member and casual rides.

Let p_M denote the electric-bike usage proportion among member rides and p_C the corresponding proportion among casual rides. We will test

$$H_0 : p_M = p_C \qquad \text{against} \qquad H_1 : p_M \neq p_C.$$

The test statistic will be

$$T = \widehat{p}_M - \widehat{p}_C.$$

We plan to conduct a permutation test by shuffling membership labels within comparable station-and-time strata where sufficient observations are available. We will compare the observed absolute difference with its permutation distribution and report both the effect size and a two-sided p-value.

We will discuss the exchangeability assumptions and dependence among rides, including the inability to account fully for repeated riders without persistent user identifiers.

As an optional extension, we will investigate whether precipitation is associated with lower daily demand after accounting for calendar and seasonal patterns. Findings will be interpreted as associations rather than causal effects.

6 Anticipated Obstacles and Challenges

- **No persistent user identifier.** We cannot follow individual riders or observe membership conversion. Conclusions will remain at the ride, station, or time-period level.

- **Unobserved demand and inventory.** Recorded trips represent completed rides rather than all attempted rentals. Historical inventory and capacity information would be needed to identify empty or full stations directly.
- **Data quality and station changes.** Missing station identifiers, inconsistent station names, changing station locations, and implausible durations may require careful cleaning.
- **Feature engineering and zero-trip hours.** We will distinguish genuine zero-trip hours from periods when stations were inactive or unavailable. We will not automatically treat every absent station-hour as zero demand.
- **Weather integration.** Observations must be aligned with trip timestamps and time zones. A limited set of weather stations may not capture local variation across the city.
- **Computational scale.** We will use efficient tools such as Polars or DuckDB, process monthly files incrementally, and consider Spark if necessary.
- **Leakage prevention and temporal changes.** Preprocessing and historical features will avoid future information. Seasonal changes may reduce performance in later months.
- **Data-use restrictions.** We will follow the published policy, avoid unauthorized branding, refrain from attempting to identify riders, and not redistribute the source data as a standalone dataset.